

Variance-based importance measures

Variance-based importance measures in the linear case

Importance measures and uncertainty quantification

- ▶ Key idea : can we quantify the amount of input uncertainty that creates dispersion in the output?
- ▶ Classical concept of dispersion : variance
- ▶ Importance of an input : amount of uncertainty (i.e., in terms of variance) it brings to the system.

Variance-based importance measures in the linear case

Importance measures and uncertainty quantification

- ▶ Key tool : Variance-based Importance Measures (VIM)
- ▶ Goal of a VIM : quantify the contribution of each input X_i to the variance of the output Y , denoted by $Var(Y)$

Variance-based importance measures in the linear case

Importance measures and uncertainty quantification

- ▶ In the linear case, VIM is usually defined from the classical variance decomposition

$$\text{Var}(Y) = \text{Var}(\mathbb{E}[Y|X]) + \mathbb{E}[\text{Var}(Y|X)]$$

- ▶ The key challenge is to decompose the so-called R^2 coefficient defined as

$$R^2 = \frac{\text{Var}(\mathbb{E}[Y|X])}{\text{Var}(Y)}$$

in a relevant manner

Variance-based importance measures in the linear case

What is a relevant decomposition for the R^2 score?
[Grömping, 2007]

Four basic desirability criteria for a VIM resulting from an R^2 decomposition:

- ▶ Proper decomposition: the sum of all weights should be equal to the total variance
- ▶ Nonnegativity: all weights should be nonnegative
- ▶ Exclusion: if $w_i = 0$, then the weight of X_i should be zero
- ▶ Inclusion: if $w_i \neq 0$, then the weight of X_i should be non zero
- ▶ (Additional) Grouping: all weights should tend to equate for highly correlated inputs.

Variance-based importance measures in the linear case

Regression coefficients and Pearson correlation for independent inputs

- ▶ Assumption : mutually independent inputs
- ▶ Law of total variance in this case

$$\text{Var}(Y) = \sum_j \beta_j^2 \text{Var}(X_j) + \sigma_\epsilon^2$$

- ▶ Hence $R^2 = \sum_j (\beta_j^*)^2$ with β_j^* standardized regression coefficient (SRC) defined as

$$\beta_j^* = \beta_j \frac{\sigma(X_j)}{\sigma(Y)}$$

Variance-based importance measures in the linear case

Regression coefficients and Pearson correlation for independent inputs

- ▶ One can see that the SRC coefficients respect the four criteria (C1), (C2), (C3) and (C4)
- ▶ If the inputs are not independent anymore, the SRC weights are no more admissible VIM
- ▶ (C1) is not respected anymore.
- ▶ Two challenges
 - ▶ Extension to non linear case?
 - ▶ Extension to non independent inputs?

Variance-based importance measures in the linear case

Roadmap

- ▶ Extension to non linear setting?
- ▶ Key ingredient : ANOVA Hoeffding decomposition
- ▶ Extension to non independent inputs?

ANOVA-Hoeffding decomposition

A vanilla example

- ▶ X_1, X_2 two discrete independent variables. X_1 (resp. X_2) uniform on $\{0, 1\}$ (resp. $\{0, 1, 2\}$)
- ▶ Let F a function defined on $\{0, 1\} \times \{0, 1, 2\}$ and $Y = F(X_1, X_2)$
- ▶ ANOVA Hoeffding decomposition of Y ?

ANOVA-Hoeffding decomposition

A vanilla example

$$Y = F_{\emptyset} + F_1(X_1) + F_2(X_2) + F_{12}(X_1, X_2)$$

with

$$F_{\emptyset} = \frac{1}{6}(F(0, 0) + F(0, 1) + F(0, 2) + F(1, 0) + F(1, 1) + F(1, 2))$$

$$F_1(x_1) = \frac{1}{3}(F(x_1, 0) + F(x_1, 1) + F(x_1, 2)) - F_{\emptyset}$$

$$F_2(x_2) = \frac{1}{2}(F(0, x_2) + F(1, x_2)) - F_{\emptyset}$$

$$F_{12}(x_1, x_2) = F(x_1, x_2) - F_1(x_1) - F_2(x_2) - F_{\emptyset}$$

ANOVA-Hoeffding decomposition

A vanilla example

Nice properties of this decomposition?

- ▶ $F_{\emptyset} = \mathbb{E}[F(X_1, X_2)]$
- ▶ $F_1(X_1) = \mathbb{E}[F(X_1, X_2)|X_1] - \mathbb{E}[F(X_1, X_2)]$
- ▶ $F_2(X_2) = \mathbb{E}[F(X_1, X_2)|X_2] - \mathbb{E}[F(X_1, X_2)]$
- ▶ $F_{12}(X_1, X_2) = F(X_1, X_2) - F_1(X_1) - F_2(X_2) - F_{\emptyset}$

ANOVA-Hoeffding decomposition

A vanilla example

Nice properties of this decomposition? **Orthogonality!**

$$\mathbb{E}[F_{\emptyset} \cdot F_1(X_1)] = 0$$

$$\mathbb{E}[F_{\emptyset} \cdot F_2(X_2)] = 0$$

$$\mathbb{E}[F_{\emptyset} \cdot F_{12}(X_1, X_2)] = 0$$

$$\mathbb{E}[F_1(X_1) \cdot F_2(X_2)] = 0$$

$$\mathbb{E}[F_1(X_1) \cdot F_{12}(X_2)] = 0$$

$$\mathbb{E}[F_2(X_2) \cdot F_{12}(X_2)] = 0$$

ANOVA-Hoeffding decomposition

Consequence

Decomposition of variance

$$\text{Var}(Y) = \text{Var}(F_{\emptyset}) + \text{Var}(F_1(X_1)) + \text{Var}(F_2(X_2)) + \text{Var}(F_{12}(X_1, X_2))$$

(global effect, marginal effects, others)

Extension to a general setting?

ANOVA-Hoeffding decomposition (1948)

Starting point : $\mathbf{X} = (X_1, \dots, X_d)$ independent random variables

Theorem [Hoeffding decomposition]

Let $F \in L^2(\mathbb{P}_X)$. Then G may be uniquely decomposed as the following orthogonal expansion

$$F(X) = \sum_{A \subset \{1, \dots, d\}} F_A(X_A) \text{ a.s.}$$

ANOVA-Hoeffding decomposition

Theorem [Hoeffding decomposition (1948)]

1. $F_{\emptyset}$ constant a.s.
2. Explicit expression of F_A

$$F_A(\mathbf{x}_A) = \sum_{B \subset A} (-1)^{|A|-|B|} \mathbb{E}[F(X) | X_B = x_B], \text{ a.s.}$$

3. For all $A \subset \{1, \dots, d\}$, for all $A \neq \emptyset$, for all $i \in A$, of F_A

$$\int_{E_i} F_A(\mathbf{x}_A) \mathbb{P}_{X_i}(dx_i) = 0.$$

ANOVA-Hoeffding decomposition

Consequence

One has

$$V = \text{Var}(F(X)) = \sum_{A \neq \emptyset} V_A$$

where

$$V_A = \sum_{B \subset A} (-1)^{|A|-|B|} \text{Var}(\mathbb{E}[F(X)|X_B])$$

Example

Assume $A = \{j\}$

$$V_j = \text{Var}(\mathbb{E}[F(X)|X_j])$$

ANOVA-Hoeffding decomposition

Exercise

- ▶ Let $X = (X_1, X_2, X_3)$ be a vector of three independent random variables
- ▶ F a square integrable random function of X_1, X_2, X_3 .
- ▶ Write the ANOVA-Hoeffding decomposition

Definition of Sobol indices

Definition (normalized indices)

One defines for a given F

- Sobol index associated to A :

$$S_A[F] := \frac{\text{Var}(F_A(X_A))}{\text{Var}(F(X))} = \frac{V_A}{V}$$

- First order Sobol index associated to the input X_j :

$$S_j[F] := \frac{\text{Var}(F_j(X_j))}{\text{Var}(F(X))} = \frac{\text{Var}(\mathbb{E}[F(X)|X_j])}{\text{Var}(F(X))}$$

- Closed Sobol index associated to A :

$$S_{A,CI}[F] := \frac{\text{Var}[\mathbb{E}(F(X)|X_A)]}{\text{Var}(F(X))}$$

Variance-based importance measures

Definition of Sobol indices

Exercise

- ▶ Let X_1, X_2, X_3 be three independent standard Gaussian variables
- ▶ Let a_1, a_2, a_3, a_{12} be four real numbers.
- ▶ Consider the following function,

$$F(x_1, x_2, x_3) = a_1 x_1 + a_2 x_2 + a_3 x_3 + a_{12} x_1 \cdot x_2$$

- ▶ Calculate the Sobol indices

Definition of Sobol indices

First properties

1. If the function F does not depend on the random variable X_i , then $S_A[F] = 0$ for any A such that $i \in A$.
2. If $T_A = 1$, then F only depends on the random variables, the indices of which are in A .
3. It is clear that

$$1 = \sum_A S_A \text{ and } 0 \leq S_A \leq 1$$

Definition of Sobol indices

Interpretation

- ▶ $S_i[F]$ quantifies the part of the variability that is due to the action of variable X_i alone on F . We speak of the **first order importance** of X_i
- ▶ $S_{i,j}[F]$ quantifies the part of the variability that is due to the interaction between the variables X_i and X_j when the first orders have been removed.

Naive approach : nested Monte Carlo estimation for Sobol' indices

What is to estimate?

- ▶ Quantity of interest : $S_i = \frac{\text{Var}_{X_i}(\mathbb{E}_{X_{-i}}[Y|X_i])}{\text{Var}(Y)}$
- ▶ For a **fixed value** of X_i , average Y over everything else (X_{-i} , i.e. all the other inputs)
- ▶ Then look at how much that averaged value **moves around** as X_i is allowed to vary
- ▶ Naive estimator : translate into two loops this definition
- ▶ Drawback : its quadratic in sample computational cost

The pick and freeze strategy

Main question

- ▶ Starting point of this strategy : **the knowledge of all closed indices** allows to recover all the other indices
- ▶ Focus on the estimation of the closed indices defined as

$$S_{(A, CI)} = \frac{\mathbb{E}(F(X)|X_A)}{\text{Var}(F(X))}$$

- ▶ Approach : the **so-called pick and freeze** strategy

The pick and freeze strategy

A simple two dimensional example

- ▶ Inputs X_1, X_2 two independent variables
- ▶ Set $Y = F(X_1, X_2)$
- ▶ Compute $S_1[F] = \text{Var}(\mathbb{E}[Y|X_1]) / \text{Var}(Y)$?

The pick and freeze strategy

A simple two dimensional example

- ▶ Let X_2' an independent copy of X_2
- ▶ Set $Y = F(X_1, X_2)$ and $Y' = F(X_1, X_2')$
- ▶ Observe that

$$\begin{aligned}\text{cov}(Y \cdot Y') &= \mathbb{E}[\mathbb{E}[F(X_1, X_2) \cdot F(X_1, X_2') | X_1]] - \mathbb{E}[F(X_1, X_2)]^2 \\ &= \mathbb{E}[\mathbb{E}[F(X_1, X_2) | X_1]^2] - \mathbb{E}[F(X_1, X_2)]^2 \\ &= \text{Var}(\mathbb{E}[F(X_1, X_2) | X_1])\end{aligned}$$

- ▶ Consequence : $S_1[F] = \text{cov}(Y \cdot Y') / \text{Var}(Y)$.

The pick and freeze strategy

First order Sobol estimator in the case of d inputs variables

- ▶ Extension to the first order Sobol indices in whole generality
- ▶ A natural estimator (Sobol 1993)

$$S_N^1 = \frac{\frac{1}{N} \sum_i Y_i Y_i' - \left(\frac{1}{N} \sum_i Y_i\right)^2}{\frac{1}{N} \sum_i Y_i^2 - \left(\frac{1}{N} \sum_i Y_i\right)^2}$$

- ▶ Improvement (Janon et al. 2014)

$$S_N^2 = \frac{\frac{1}{N} \sum_i Y_i Y'_i - \left(\frac{1}{N} \sum_i \left[\frac{Y_i + Y'_i}{2} \right] \right)^2}{\frac{1}{N} \sum_i \left[\frac{Y_i^2 + Y_i'^2}{2} \right] - \left(\frac{1}{N} \sum_i \left[\frac{Y_i + Y'_i}{2} \right] \right)^2}$$

- ▶ LLN and CLT hold

The pick and freeze strategy

Total effect estimation

- ▶ Let X'_j an independent copy of X_j
- ▶ In the same way, set $\tilde{Y} = F(X'_j, X^{(-j)})$

$$\text{cov}(Y, \tilde{Y}) = \text{Var}(\mathbb{E}[Y|X^{(-j)}])$$

- Then

$$\begin{aligned} S^{(-j)} &= 1 - \text{Var}(\mathbb{E}[Y|X^{(-j)}]) / \text{Var}(Y) \\ &= (\text{Var}(Y) + \text{Var}(\tilde{Y}) - 2 \cdot \text{Cov}(Y, \tilde{Y})) / (2 \cdot \text{Var}(Y)) \\ &= \mathbb{E}[(Y - \tilde{Y})^2] / (2 \cdot \text{Var}(Y)) \end{aligned}$$

The pick and freeze strategy

Total effect estimation

- ▶ $(X^{(A)}, X^{(B)})$ two copies of the input variables
- ▶ Set $Y^{(A)} = F(X^{(A)})$, $Y^{(BA)} = F(X_j^{(B)}, X^{(A,-j)})$
- ▶ A natural estimator can be introduced as in the case of first order Sobol indices

Case of dependent inputs

Extension of to more general case (Borganavo et al., 2024)

- ▶ Case of dependent inputs?
- ▶ Assume that for some j , $X^{(-j)}$ and $X'^{(-j)}$ are independent conditionally to X_j
- ▶ The equality

$$\text{cov}(Y, Y') = \text{Var}(\mathbb{E}[F(X, X')|X])$$

still holds but the way we compute this expectation involves a density ratio

Given-data estimation: no paired design at all

Gamboa, Gremaud, Klein & Lagnoux (2020), building on Chatterjee's (2021) coefficient of correlation: sort by X_i , measure how much neighbouring Y 's still "stick together."

$$\hat{S}_i = \frac{\frac{1}{n} \sum_{j=1}^{n-1} Y_{(j)} Y_{(j+1)} - \bar{Y}^2}{\frac{1}{n} \sum_j Y_j^2 - \bar{Y}^2}$$

Result ($n = 5000$, a *single* existing sample)

$$\hat{S} = (0.3081, 0.4606, -0.0168) \quad \text{vs. true} \\ (0.3139, 0.4424, 0.0000)$$

Needs only N evaluations, not $(d+2)N$ — and works on data you already have, with no special paired design required.

Références

- ▶ Sobol, I.M. (1993) — Sensitivity analysis for non-linear mathematical models
- ▶ Saltelli, A. (2002) — Making best use of model evaluations to compute sensitivity indices
- ▶ Jansen, M.J.W. (1999) — Analysis of variance designs for model output

Références

- ▶ Janon, Klein, Lagnoux, Nodet & Prieur (2014) — Asymptotic normality and efficiency of two Sobol index estimators
- ▶ Owen, A.B. (2014) — Sobol' indices and Shapley value

Références

- ▶ Gamboa, Gremaud, Klein & Lagnoux (2022) — Global Sensitivity Analysis: a new generation of mighty estimators based on rank statistics
- ▶ Chatterjee, S. (2021) — A new coefficient of correlation
- ▶ Kucherenko, Tarantola & Annoni (2012) — Estimation of global sensitivity indices for models with dependent variables
- ▶ Borgonovo, E., Plischke, E., Prieur, C. (2024). Total effects with constrained features. *Statistics and Computing*, 34(2), 87

Details about naive approach : nested Monte Carlo estimation for Sobol' indices

Step 1 : estimation of $\mathbb{E}[Y \mid X_i = x_i]$ for a fixed i

- ▶ Fix X_i at some drawn value $x_i^{(k)}$.
- ▶ Draw N_{inner} samples of **all the other inputs** X_{-i} (independently, from their own distributions), keeping $X_i = x_i^{(k)}$ fixed throughout.
- ▶ Run the model on each of these N points, and **average the outputs**:

$$\widehat{\mathbb{E}}[Y \mid X_i = x_i^{(k)}] = \frac{1}{N_{\text{inner}}} \sum_{j=1}^{N_{\text{inner}}} F(x_i^{(k)}, X_{-i}^{(j)})$$

Details about naive approach : nested Monte Carlo estimation for Sobol' indices

Step 2 : repeat for many values of X_i , then look at the spread
 textbfPurpose: see how much that inner-average moves as X_i itself changes — this spread *is* the variance being sought.

- ▶ Draw N_{outer} different values of X_i : $x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(N_{\text{outer}})}$.
- ▶ For **each one**, run the entire inner loop above to get one inner-average $\hat{\mathbb{E}}[Y | X_i = x_i^{(k)}]$.
- ▶ With N_{outer} inner-averages in hand — one per outer draw — take the **variance of these numbers**:

$$\hat{S}_i = \frac{\widehat{\text{Var}}_k \left(\hat{\mathbb{E}}[Y | X_i = x_i^{(k)}] \right)}{\widehat{\text{Var}}(Y)}$$

Details about naive approach : nested Monte Carlo estimation for Sobol' indices

Ground truth: the Ishigami function

$$F(x_1, x_2, x_3) = \sin(x_1) + a \sin^2(x_2) + b x_3^4 \sin(x_1), X_i \sim \text{Uniform}(-\pi, \pi)$$

with the classic parameters $a = 7$, $b = 0.1$.

Details about naive approach : nested Monte Carlo estimation for Sobol' indices

Computational cost of estimating S_1 in the naive way

$N_2 = 4$ and $N_1 = 3$:

Outer draw k	$x_1^{(k)}$ (fixed)	Inner draws of (x_2, x_3)	Inner average
1	0.5	3 (x_2, x_3) pairs	$\bar{y}_1$
2	1.8	3 <i>different</i> pairs	$\bar{y}_2$
3	-0.2	3 more pairs	$\bar{y}_3$
4	2.9	3 more pairs	$\bar{y}_4$

Total model evaluations: $4 \times 3 = 12$ (in general $N_{\text{outer}} \times N_{\text{inner}}$)